



QUANTITATIVE GENETICS FOR PLANT BREEDING

CONTENTS



Chapter 11: Multiple Trait Selection

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Most cultivar improvement programs involve selection for multiple traits. Depending on the program or project goals, one of three types of multiple-trait selection can be employed. A brief description of these types follows.

- **Multistage selection:** Selection for different traits at different stages during cultivar development.
- **Tandem selection:** Selection for one trait until that trait is improved, then for a second, etc., until finally each has been improved to the desired level.
- **Independent culling levels:** A certain level of merit is established for each trait, and all individuals below that level are discarded regardless of values for other traits.
- **Index selection:** Select for all traits simultaneously by using some index of net merit.

Learning Objectives

- Explain the role of selection for multiple traits on genetic gain.
- Learn methods for selecting multiple traits.
- Learn how to evaluate the efficiency and effectiveness of selection on multiple traits.

Index Selection

An **index** is the best linear prediction of an individual's or line's breeding value and takes the form of the multiple regression of breeding value on all sources of information.

The objective of an index is to find the linear combination of phenotypic values that maximizes the expected gain or, equivalently, that maximizes the correlation between the index value and true worth (breeding value).

Index Selection Theory

Define phenotype as affected by components in Equation 1

$$P_i = G_i + E_i (i = 1, \dots, n)$$

Equation 1 Linear model for sources of variability in phenotype.

where:

P_i = the observed value of the attribute i for an individual or line,

G_i = the average of phenotypic values over a population of environments,

E_i = non-genotypic contributions from environments.

Note: Genotype x environment interactions are permitted, assuming genotypes and environments are associated entirely at random; such interactions are incorporated into E_i (Equation 1). If GxE are not random, then see Cooper and DeLacy (1994).

Assume G_i is composed entirely of additive effects of genes (breeding values). Define the genotypic economic value, H , of an individual as written in Equation 2:

$$H = \sum_{i=1}^n a_i G_i$$

Equation 2 Linear model for sources of variability in phenotype.

where:

a_i = known relative economic values.

Assume the quantities P_i and H are such that the regression of P_i on H is linear. Selection will then be based on the linear function, I (Equation 3).

$$I = \sum_{i=1}^n b_i P_i = \sum_{i=1}^n b_i (G_i + E_i)$$

Equation 3 Linear model for index selection.

where:

b_i = regression coefficient,

P_i, G_i, E_i are as defined previously.

Assumptions

Assume an equal amount of information on all individuals to be evaluated and selected. Also, assume that the distributions of P_i , G_i , and E_i are unknown but that the mean and covariances are known.

Then,

$$P_i \sim NID(0, \sigma_{ii}^P) (i = 1, \dots, n) \quad G_i \sim NID(0, \sigma_{ii}^G) (i = 1, \dots, n) \quad E_i \sim NID(0, \sigma_{ii}^E) (i = 1, \dots, n) \quad Cov(G_i, E_i) = 0 (i = 1, \dots, n)$$

Mean and Covariance of H and I

With these assumptions, we can derive the mean and covariance of H and I as in the set of equations written in Equation 4.

$$\begin{aligned} E(H) &= \sum_{i=1}^n a_i E(G_i) = 0, & V(H) &= \sigma_H^2 = \sum_{i=1}^n \sum_{j=1}^n a_i a_j \sigma_{ij}^G \\ E(I) &= \sum_{i=1}^n b_i E(P_i) = 0, & V(I) &= \sigma_I^2 = \sum_{i=1}^n \sum_{j=1}^n b_i b_j \sigma_{ij}^P \\ Cov(I, H) &= \sigma_{IH} = \sum_{i=1}^n \sum_{j=1}^n a_i b_j \sigma_{ij}^G, & Cov(G_i, I) &= \sigma_{GiL} = \sum_j b_j \sigma_{ij}^G \end{aligned}$$

Equation 4 Formulae for estimating mean, variance, and covariance of H , I , and G_i .

where:

terms are as defined previously.

The objective of a selection index is to use some linear combination of trait values (I) to predict true genetic worth (H).

- maximizing expected genetic gain.
- maximizing the correlation of the sample index (I) with true worth (H).
- maximizing the probability of correct selection.
- minimizing the $E(I-H)^2$.

Williams (1962) showed that maximizing the correlation between I and H also maximizes the expected genetic gain and the probability of correct selection.

Derivation of the Optimum Index

Maximizing correlation of I with H (Equation 5):

$$r_{IH} = \frac{Cov(I,H)}{\sqrt{V(I)V(H)}} = \frac{\sum_i \sum_j a_i b_j^G \sigma_{ij}}{\sqrt{(\sum_i \sum_j b_i b_j^G \sigma_{ij})(\sum_i \sum_j a_i a_j^G \sigma_{ij})}}$$

Equation 5 Equation for estimating the correlation between I and H.

where:

terms are as defined in equation 4.

It can be shown that maximizing r_{IH} is equivalent to maximizing $\log(r_{IH})$ (Equation 6).

$$\log r_{IH} = \log \left(\sum_i \sum_j a_i b_j^G \sigma_{ij} \right) - \frac{1}{2} \log \left(\sum_i \sum_j b_i b_j^G \sigma_{ij} \right) - \frac{1}{2} \log \left(\sum_i \sum_j a_i a_j^G \sigma_{ij} \right)$$

Equation 6 Equation for maximizing correlation of I with H.

where:

terms are as defined previously.

Using least squares and differentiating with respect to b_j , we get Equation 7:

$$\frac{\partial \log(r_{IH})}{\partial b_j} = \frac{\sum_i a_i^G \sigma_{ij}}{\sigma_{IH}} - \frac{1}{2} \frac{2 \sum_i b_i^G \sigma_{ij}}{\sigma_{ij}^2} = 0 \quad (j = 1, \dots, n), \quad \frac{\sum_i b_i^G \sigma_{ij}}{\sigma_{ij}^2} = \frac{\sum_i a_i^G \sigma_{ij}}{\sigma_{IH}} \quad (j = 1, \dots, n)$$

Equation 7 Calculating the correlation of I with H using least squares techniques.

where:

all terms are as defined previously.

Rewriting the Normal Equations

These equations are called the normal equations (unrelated to normal distribution) and constitute n equations in n unknowns.

They can be rewritten as in Equation 8:

$$\sum_i b_i^G \sigma_{ij} = \sum_i a_i^G \sigma_{ij} \left(\frac{\sigma_{II}^2}{\sigma_{IH}^2} \right) \quad j = (1, \dots, n)$$

Equation 8 Calculating the regression coefficient and observed value.

where:

all terms are as defined previously.

Because we are only interested in relative values of b_i , the constant term can be dropped, resulting in Equation 9:

$$\sum_i b_i^G \sigma_{ij} = \sum_i a_i^G \sigma_{ij} \quad j = (1, \dots, n)$$

Equation 9 Dropping the constant values before calculating the regression coefficient and observed value.

all terms are as defined previously.

Further Calculations

Considering 2 traits ($n=2$), Equation 9 is written as:

$$b_1^P \sigma_{11} + b_2^P \sigma_{21} = a_1^G \sigma_{11} + a_2^G \sigma_{21} \quad (j = 1); \quad b_2^P \sigma_{12} + b_2^P \sigma_{22} = a_1^G \sigma_{12} + a_2^G \sigma_{22} \quad (j = 2)$$

Solving the above equations, we get Equation 10:

$$b_1 = \frac{a_1[G\sigma_{11}^P \sigma_{22} - G\sigma_{12}^P \sigma_{21}] + a_2[G\sigma_{21}^P \sigma_{22} - G\sigma_{22}^P \sigma_{21}]}{P\sigma_{11}^P \sigma_{22} - P\sigma_{12}^P \sigma_{21}},$$

$$b_2 = \frac{a_1[G\sigma_{12}^P \sigma_{11} - G\sigma_{11}^P \sigma_{12}] + a_2[G\sigma_{22}^P \sigma_{11} - G\sigma_{21}^P \sigma_{11}]}{P\sigma_{11}^P \sigma_{22} - P\sigma_{12}^P \sigma_{21}}.$$

Equation 10 Calculating the regression coefficient,

where:

all terms are as defined previously.

Minimizing $E(I-H)^2$

The same equations can be derived to get Equation 9 by minimizing $E(I-H)^2$ as written below:

$$E(I-H)^2 = E\left[(\sum_i b_i P_i) - (\sum_i a_i G_i)\right]^2 = E(\sum_i b_i P_i)^2 - 2E\left[(\sum_i b_i P_i)(\sum_i a_i G_i)\right] + E(\sum_i a_i G_i)^2$$

because,

$$E(\sum_i b_i P_i)^2 = V(I); \quad E\left[(\sum_i b_i P_i)(\sum_i a_i G_i)\right] = Cov(I, H); \quad E(\sum_i a_i G_i)^2 = V(H)$$

Applying least squares:

$$\frac{\partial E(I-H)^2}{\partial b_j} = 2 \sum_i b_i^P \sigma_{ij} - 2 \sum_i a_i^G \sigma_{ij} = 0 \quad (j = 1, \dots, n)$$

Dividing through by 2 and rearranging, we get the normal equations:

$$\sum_i b_i^P \sigma_{ij} = \sum_i a_i^G \sigma_{ij} \quad j = (1, \dots, n)$$

which are identical to the equations presented previously.

Expected Genetic Gains

To derive the expected genetic gains, we need to make assumptions about the distributions of P_i , G_i , and E_i .

Assume:

1. P_i , G_i , and E_i are distributed normally with the mean and covariance structure given earlier.
2. Truncation selection on I .

Then, the expected genetic gain is estimated with Equation 11:

$$\Delta H = E(\bar{H}_S - \bar{H}) = \beta_{HI}(\bar{I}_S - \bar{I})$$

Equation 11 Calculating the genetic gain of change in genetic worth,

where:

ΔH = the genetic gain, that is, the change in genetic worth,

β_{HI} = the regression coefficient of H on I , which gives the mean value of H or any value of I .

This is the standard way to calculate predicted gains from univariate selection. See, for example, Empig et al. (1972).

Truncation Selection

The situation with regard to truncation selection is based on the following where,

$\bar{I}$ is the mean value of the index in the population; c is the truncation point; z is the height of ordinate of the standard normal curve at the truncation point c ;

P is the proportion of the population selected, and $\bar{I}_S$ is the mean of the selected individuals.

Then, the value of the frequency of the index is estimated with Equation 12:

$$f(I) = \frac{1}{\sigma_I \sqrt{2\pi}} e^{-\left[\frac{(I-\bar{I})^2}{2\sigma^2}\right]}$$

Equation 12 Calculating the genetic gain of change in genetic worth,

where:

$f(I)$ = the frequency of individuals with index value I

other terms are as defined previously.

Selection Relationships

The proportion saved is related to the truncation point by $S = \int_c^\infty f(I) dI$, and the mean value of the selected group is $\bar{I}_S = \frac{1}{S} \int_c^\infty I f(I) dI$.

The selection differential (D) is given by Equation 13:

$$D = (\bar{I}_S - \bar{I}) = \frac{1}{S} \left[\int_c^\infty I f(I) dI - \bar{I} \right] = \frac{1}{S} \int_c^\infty (I - \bar{I}) f(I) dI$$

Equation 13 Formula for calculating the selection differential, D ,

where:

S, I are as defined previously.

Standardized Regression Coefficient

Let,

$$i = \frac{I - \bar{I}}{\sigma_I}; \quad c = \frac{C - \bar{I}}{\sigma_I}; \quad \partial i = \frac{\partial I}{\sigma_I};$$

then, the standardized regression coefficient is derived as in Equation 14:

$$DD = \frac{\sigma_I}{S} \frac{1}{\sqrt{2\pi}} \int_c^\infty i e^{-\left[\frac{i^2}{2}\right]};$$

$$\partial i = \frac{\sigma_I}{S} \frac{1}{\sqrt{2\pi}} e^{-\left[\frac{c^2}{2}\right]};$$

$$D = \frac{\sigma_i}{s} z$$

Equation 14 Calculating the standardized regression coefficient,

where:

z = the height of the ordinate at the truncation point.

Typically, this is represented as $\frac{S}{\sigma_I} = \frac{z}{P} = i$, where, k is the standardized regression coefficient.

Expected Gain

Expected gain can then be represented as in Equation 15:

$$\Delta H = kb_{HI}\sigma_I,$$

Equation 15 Formula for expected genetic gain,

where:

$$b_{HI} = \frac{Cov(H,I)}{\sigma_I^2},$$

$$Cov(H,I) \sum_i \sum_j a_i b_j^G \sigma_{ij} = \sum_i b_j \sum_i a_i^G \sigma_{ij}$$

From the normal equations derived earlier, we have $\sum_i b_i^P \sigma_{ij} = \sum_i a_i^G \sigma_{ij} \quad j = (1, \dots, n)$

Substituting these terms,

$Cov(H,I) = \sum_j b_j \sum_i b_i^P \sigma_{ij} = \sum_i \sum_j b_i b_j^P \sigma_{ij}$; $Cov(H,I) = \sigma_I^2$, we get expected genetic as written in Equation 16:

$$\Delta H = \frac{kCov(H,I)}{\sigma_I^2} \sigma_I = k \frac{\sigma_I^2}{\sigma_I^2} \sigma_I = k\sigma_I,$$

Equation 16 Alternative formula for expected genetic gain,

where:

terms are as defined previously.

Predicted Gain

The predicted gain is more useful when written in terms of the correlation between **H** and **I** designated as r_{HI} (Equation 17):

$$\Delta H = \frac{kr_{IH}\sigma_H\sigma_I^2}{\sigma_I^2} = kr_{IH}\sigma_H,$$

Equation 17 Formula for predicted genetic gain,

where:

$$r_{HI} = \frac{Cov(H,I)}{\sigma_I\sigma_H} = \frac{\sigma_I}{\sigma_H}, \text{ as defined previously.}$$

In the selection index literature, r_{HI} is called the **accuracy of selection** because it is a measure of how well the index, **I**, measures the true worth, **H**.

Alternative selection indices, **I**, can be compared using r_{HI} as long as the selection goal, **H**, remains the same for each of the indices.

Expected Genetic Gains for Each Trait

Let ΔG_i be the expected genetic gain in trait *i* when selection is on **I**. From representations in previous equations as below, the expected genetic gains can be obtained as written in Equation 18.

$$\Delta H = \sum_i a_i \Delta G_i,$$

Equation 18 Formula for expected genetic gains for each trait,

where:

$$\Delta G_i = kb_{GiI}\sigma_I (i = 1, \dots, n) = k \frac{Cov(G_i,I)\sigma_I}{\sigma_I^2} \left(\frac{\sigma_{GiI}}{\sigma_{GiI}} \right) = k \frac{Cov(G_i,I)}{\sigma_I \sigma_{GiI}} \sigma_I \sigma_{GiI} = kr_{GiI} \sigma_{GiI},$$

$$\begin{aligned} \text{Cov}(G_i, I) &= E[G_i, \sum_j b_j P_j] = \sum_j b_j G \sigma_{ii}, \\ \sum_i a_i \text{Cov}(G_i, I) &= \sum_i \sum_j a_i b_j G \sigma_{ij} = \text{Cov}(H, I). \end{aligned}$$

This index requires that you know the true values of the population parameters. However, estimates of the population parameters are often substituted for the true values, and the resulting index is called the **estimated index** or the **Smith-Hazel index**.

Matrix Representation of Selection Indices

With this notation, the normal equations can be written as $Pb = Ga$, and $b = P^{-1}Ga$.

Some Results

$$\sigma_I^2 = \underline{b}'P\underline{b}; \quad \sigma_H^2 = \underline{a}'G\underline{a}; \quad \sigma_{IH} = \underline{b}'Ga = \underline{b}'Pb = \sigma_I^2; \quad \text{Cov}(G_i, I) = \sigma_{GiI} = \underline{b}'G_i,$$

where G_i is the i th row of G . The genetic gain can be written as in Equation 19:

$$\Delta H = k\sigma_I = k\sqrt{\underline{b}'P\underline{b}}; \quad \underline{\Delta} = \begin{pmatrix} \Delta G_1 \\ \Delta G_2 \\ \vdots \\ \Delta G_n \end{pmatrix}; \quad \Delta = \frac{Gb}{\sqrt{\underline{b}'P\underline{b}}}; \quad \Delta H = \underline{a}' \underline{\Delta},$$

Equation 19 Matrix notation for expected genetic gains,

where:

$$r_{IH} = \sqrt{f \frac{\underline{b}'Ga}{\underline{a}'Ga}},$$

other terms are as defined previously.

Construction of a Selection Index

Optimum Index

The normal equations can be written for an optimum index as $P\underline{b} = G\underline{a}$; $\underline{b} = P^{-1}G\underline{a}$, where P , G , and $\underline{a}$ are known without error, and the index is as in Equation 20:

$$I = \underline{b}'\underline{x}; \quad H = \underline{a}'\underline{y},$$

Equation 20 Formula for optimum index,

where:

$\underline{x}$ = trait x ,

$\underline{y}$ = trait y .

Smith-Hazel Index

This index is the same as the optimum index; only in this case (Equation 21), we use estimates of P , G , and $\underline{a}$, designated as $\hat{P}$, $\hat{G}$, and $\hat{\underline{a}}$, respectively.

$$\hat{P}\hat{\underline{b}} = \hat{G}\hat{\underline{a}}; \quad \hat{\underline{b}} = \hat{P}^{-1}\hat{G}\hat{\underline{a}}; \quad \hat{I} = \hat{\underline{b}}'\hat{\underline{x}}; \quad H = \hat{\underline{a}}'\hat{\underline{y}},$$

Equation 21 Formula for Smith-Hazel index,

where:



Base Index

The Base Index was apparently first suggested by Brim et al. (1959) and named the Base Index by Williams (1962). The base index is constructed simply by allowing $\underline{b} = \underline{a}$ and written as in Equation 22, with all terms defined earlier.

$$I = \underline{a}'\underline{x}; \quad H = \underline{a}'\underline{y},$$

Equation 22 Formula for Smith-Hazel index,

Some results for this index include:

$$\sigma_I^2 = \underline{a}'\underline{P}\underline{a}; \quad \sigma_H^2 = \underline{a}'\underline{a}; \quad \text{Cov}(I, H) = \underline{a}'\underline{G}\underline{a}; \quad r_{IH} = \frac{\underline{a}'\underline{G}\underline{a}}{\sqrt{(\underline{a}'\underline{P}\underline{a})(\underline{a}'\underline{G}\underline{a})}} = \sqrt{\frac{\underline{a}'\underline{G}\underline{a}}{\underline{a}'\underline{P}\underline{a}}}$$

The foremost attribute of this index is its simplicity of construction and interpretation. Also, this index does not require the estimation of genetic parameters.



Multiplicative Index

The multiplicative index was first proposed by Elston (1963). This index is also sometimes called the weight-free index because it does not require the specification of index weights or economic values.

The general form of this index is as in Equation 23:

$$I = (X_1 - k_1)(X_2 - k_2) \cdots (X_n - k_n),$$

Equation 23 Formula for multiplicative index,

where:

k_1 = the minimum value of trait X_1 set by the breeder.

In addition to being weight-free, this index also does not require the estimation of genetic or phenotypic parameters. Because this is a curvilinear index, theory is not available to predict gains. Baker (1974) found that this index can be approximated by using a linear index, where the weights are the reciprocals of the phenotypic standard deviations of the traits in the index. This essentially amounts to an index with equal weighting per phenotypic standard deviation. Approximate, predicted gains can then be obtained for this index using the Smith-Hazel index theory.

Desired Gain Index

The desired gain index was suggested by Pesek and Baker (1969). This index allows the breeder to specify a vector of desired gains, $\underline{q}$, and then substitute this into the predicted gain equation and solve for the index weights. The solution for $\underline{b}$ is as in Equation 24:

$$\underline{b} = G^{-1}\underline{q},$$

Equation 24 Formula for desired gain index,

where:

$\underline{q}$ = the vector of desired gains,

$I = \underline{b}'\underline{x}$ is as defined previously.

This index was proposed to eliminate the need to specify economic weights. However, in practice, there are some difficulties with the index in specifying the vector of desired gains.

This index will result in maximum gains in each trait according to the relative importance assigned by the breeder in specifying the desired gains.



equations.

Restricted Selection Index

Restricted selection indices were first derived by Kempthorne and Nordskog (1959). Since then, various restricted indices have been derived by Cunningham et al. (1970) and James (1968). See Lin (1978) for a complete list. Basically, restricted selection indices involve holding the genetic gains in one or more traits to a constant or zero while changing the means of other traits in the desired direction. The basic method is to impose the restriction on the index equations that $Cov(G_i, I) = 0$.

The simplest procedure to accomplish this was given perhaps by Cunningham et al. (1970). Their method involved solving the following set of equations in Equation 25:

$$\begin{pmatrix} P & G_i \\ G_i & O \end{pmatrix} \begin{pmatrix} \underline{b} \\ \underline{b_d} \end{pmatrix} = \begin{pmatrix} G \\ O \end{pmatrix} [a],$$

Equation 25 Formula for restricted selection index,

where:

$$\underline{b} = [I - P^{-1}G_i(G_i'P^{-1}G_i)^{-1}G_i']P^{-1}Ga,$$

$$\underline{b_d} = (G_i'P^{-1}G_i)^{-1}G_i'P^{-1}Ga,$$

other terms are as defined previously.

$\underline{b}$ is the vector of index weights to use in the index equation as $I = \underline{b}'x$, as before. The dummy variable is not used in the index equation.

This method has the interesting consequence that the value obtained for the dummy variable is the negative of the economic weight needed to produce zero change in that trait in an unrestricted selection index.

Rank Summation Index

The rank summation index was first suggested by Mulamba and Mock (1978). Basically, this index involves obtaining the ranks of each of the traits to be included in the index and then calculating the index by summing up the trait ranks, represented in Equation 26.

$$I = \sum_{i=1}^n \text{rank}(X_i).$$

Equation 26 Formula for the rank summation index,

where:

terms are as defined previously.

The primary advantages of this index are that genetic parameters need not be calculated, it transforms the data so that the variances for each trait are identical, and it does not require the specification of economic weights, although they can be used.

As with the multiplicative index, predicted gains cannot be calculated for this index. However, Crosbie et al. (1980) found that the same prediction equation used for the multiplicative index provides a reasonably good approximation of the predicted gains for the rank summation index.

Selection Index Efficiency

Methods to Compare Selection Index Efficiency

Cunningham (1969) provided a method for comparing the relative efficiencies of selection indices. He was primarily interested in deleting traits from the index so that their relative contribution to the gain in the true worth (H) could be determined. Dropping

Define the index containing all the traits of interest as the original index and define the index with one trait dropped out as the i^{th} reduced index. Then the efficiency of the i^{th} reduced index relative to that of the original index is the ratio of their standard deviations. Cunningham showed this to be as in Equation 27:

$$\sqrt{\frac{b'Pb - \frac{b_i^2}{W_{ii}}}{b'Pb}}$$

Equation 27 Formula for selection index efficiency,

where:

b_i = the i^{th} weighting factor in the original index,

W_{ii} = the corresponding diagonal element in the inverse of P.

A more usual procedure is to compare the gain for the i^{th} trait when selection is on I, relative to the single trait selection for the i^{th} trait.



Effect of Correlations on Index Weights

To determine the effects of correlation on index weights, we need to derive the index equations $P\underline{b} = G\underline{a}$ in terms of genetic and phenotypic correlation coefficients (Equations 28, 29, 30) following the series of derivations as follows. Let,

$$X_i^* = \frac{X_i}{\sqrt{P\sigma_{ii}}} = \frac{G_i + E_i}{\sqrt{P\sigma_{ii}}} \quad (i = 1, \dots, n) \text{ Then: } V(X_i^*) = P\sigma_{ii} = 1 \text{Cov}(X_i^*, X_j^*) = \frac{P\sigma_{ij}}{\sqrt{P\sigma_{ii}P\sigma_{jj}}} = P r_{ij} \quad (i = j = 1, \dots, n) V(G_i^*)$$

Derivation

$$P^* = \begin{pmatrix} 1 & P r_{12} & \dots & P r_{1n} \\ P r_{21} & 1 & \dots & P r_{2n} \\ \vdots & \vdots & & \vdots \\ P r_{n1} & P r_{n2} & \dots & 1 \end{pmatrix}; \quad G^* = \begin{pmatrix} h_{11} & h_{12} & \dots & h_{1n} \\ h_{21} & h_{22} & \dots & h_{2n} \\ \vdots & \vdots & & \vdots \\ h_{n1} & h_{n2} & \dots & h_{nn} \end{pmatrix}; \quad \hat{\underline{b}}^* = (P^*)^{-1} G^* \underline{a}$$

Equation 28 Formula for phenotypic and genotypic correlations,

where:

terms are as defined previously.

For 2 traits (n=2),

$$\hat{\underline{b}}^* = \begin{pmatrix} a_1(h_{11}^2 - P r_{12}h_{12}) + a_2(h_{12} - P r_{12}h_2^2) \\ a_1(h_{12} - P r_{12}h_1^2) + a_2(h_2^2 - P r_{12}h_{12}) \end{pmatrix} \left(\frac{1}{1 - (P r_{12})^2} \right)$$

Equation 29 Formula for estimated correlation between two traits,

where:

terms are as defined previously.

When correlations are zero: $P r_{12} = G r_{12} = 0$,

$$\hat{\underline{b}}^* = \begin{pmatrix} a_1 h_1^2 \\ a_2 h_2^2 \end{pmatrix}; \quad P = \begin{pmatrix} P\sigma_{11} & 0 \\ 0 & P\sigma_{22} \end{pmatrix}; \quad P^{-1} = \begin{pmatrix} \frac{1}{P\sigma_{11}} & 0 \\ 0 & \frac{1}{P\sigma_{22}} \end{pmatrix}; \quad G = \begin{pmatrix} G\sigma_{11} & 0 \\ 0 & G\sigma_{22} \end{pmatrix}$$

Equation 30 Formula for estimated correlation between two traits when phenotypic and genetic correlations are zero,

where:

terms are as defined previously.



When $|r| < 0.30$, the use of the above index is nearly as efficient as using the Smith-Hazel index.

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